

PUBLISHED: 28-APR-2017
2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

RIGHT CYLINDER HEAD (G1509345)

REMOVAL AND INSTALLATION

12.29.05	CYLINDER HEAD - ASSEMBLY - RIGHT - RENEW	5000 CC, AJ V8, SUPERCHARGED	18.8	USED WITHINS
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SPECIAL TOOL(S)



PART(S)

STEP	PART NAME	QUANTITY
Installation Step 2	Cylinder head gasket	1
Installation Step 3	Cylinder head bolts	1
Installation Step 11	Oil filter housing gasket	1
Installation Step 12	Oil filter O-ring seal(s)	1
Installation Step 14	Cylinder block coolant outlet O-ring seals	1

REMOVAL

NOTES:

- Removal steps in this procedure may contain installation details.
- Some variation in the illustrations may occur, but the essential information is always correct.

1. Disconnect the battery ground cable.

Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).

- 2.



WARNING:

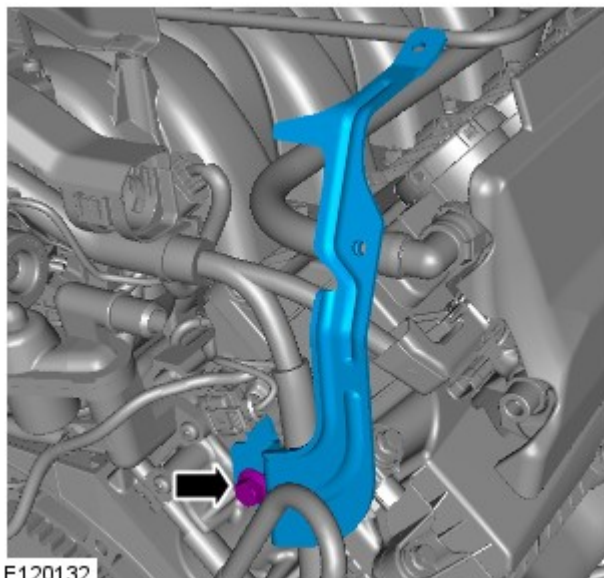
Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

3. Refer to: Right Camshaft (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).

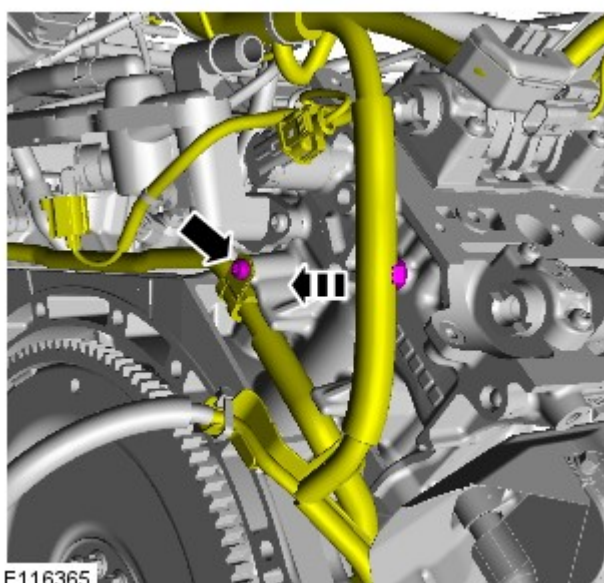
4. Refer to: Right Exhaust Manifold (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).

- 5.



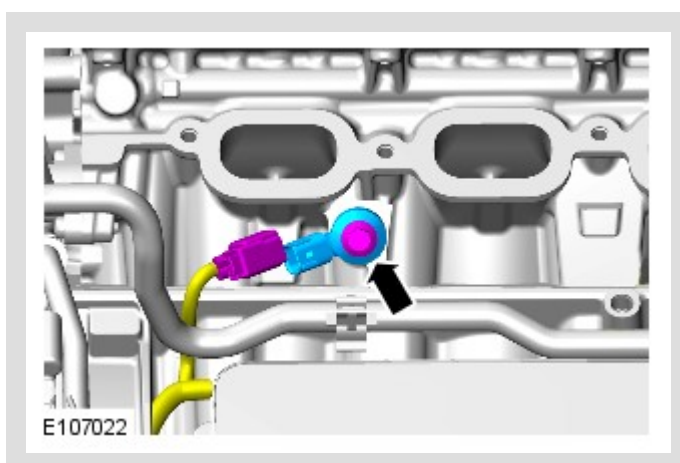
E120132

6.

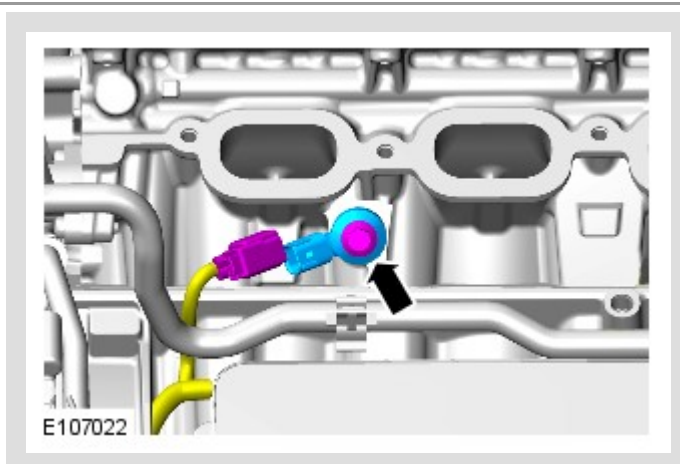


E116365

7.



8.

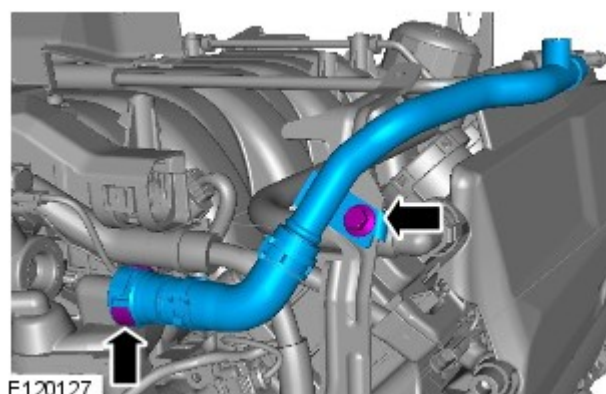


9.



CAUTION:

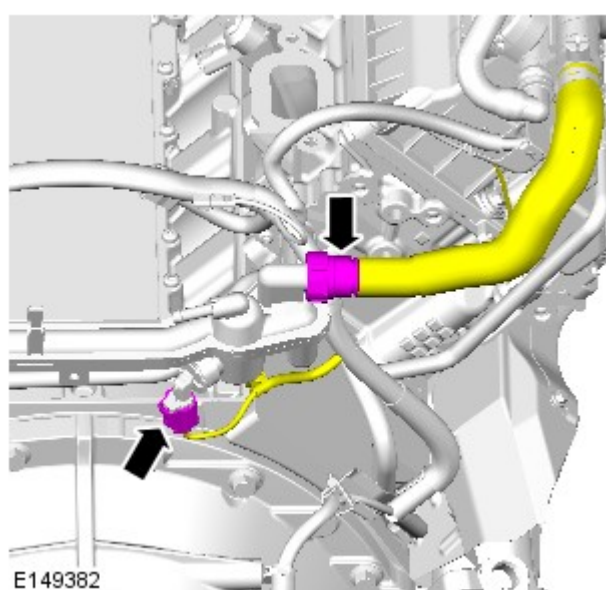
Be prepared to collect escaping coolant.



10.

⚠ CAUTION:

Be prepared to collect escaping coolant.

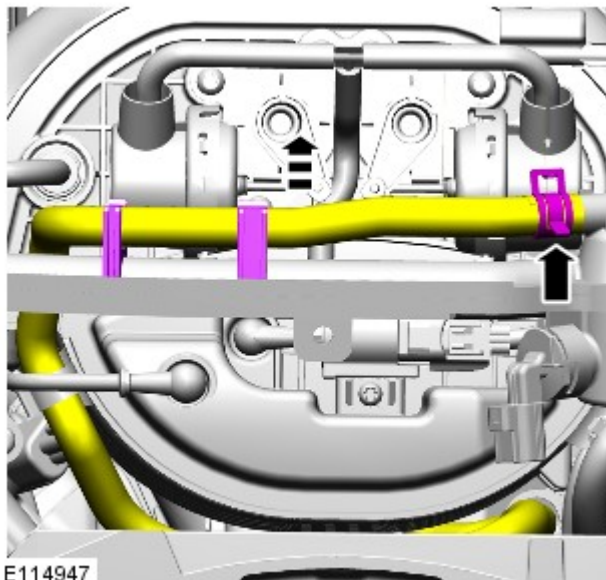


11.



NOTE:

Clamp the hose to minimize coolant loss.

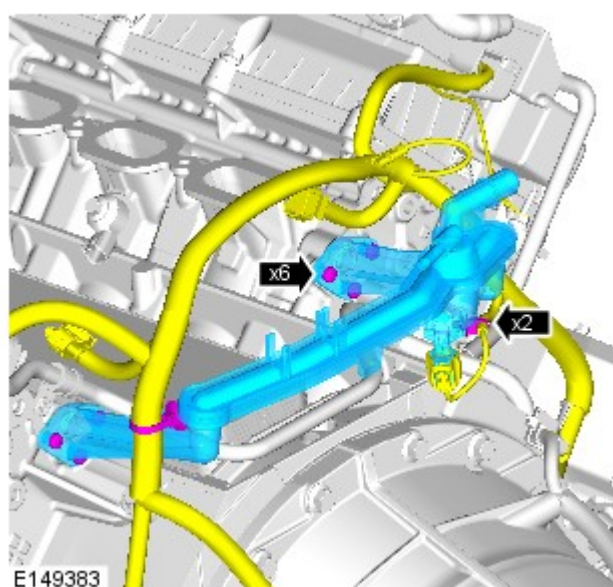


12.

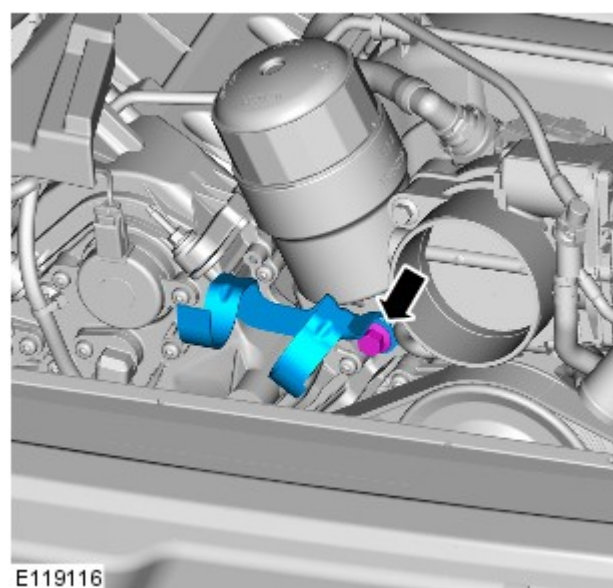


CAUTION:

Discard the seals.



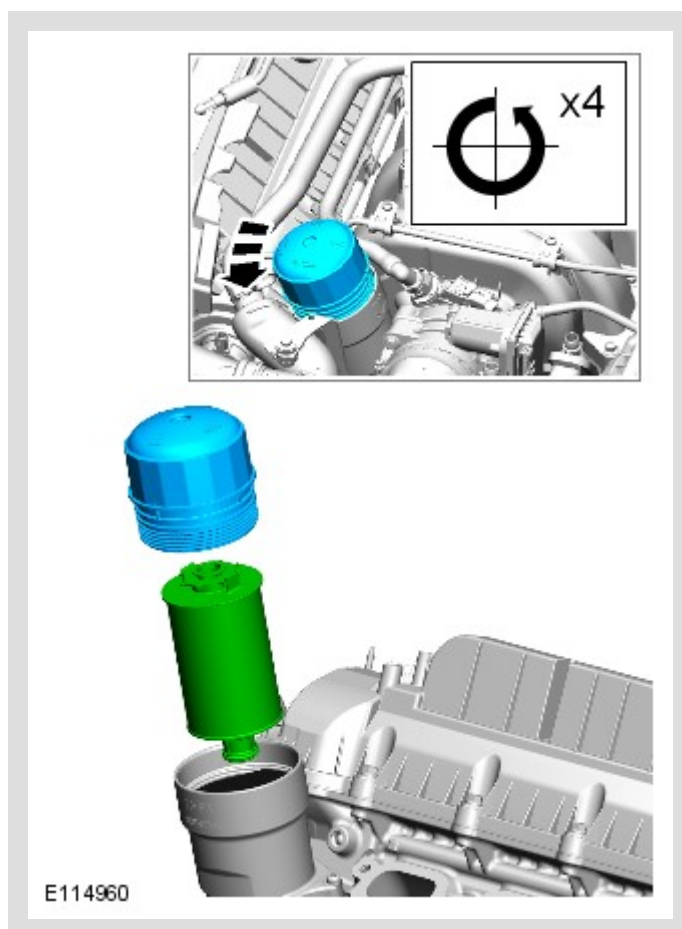
13.



14.

△ NOTE:

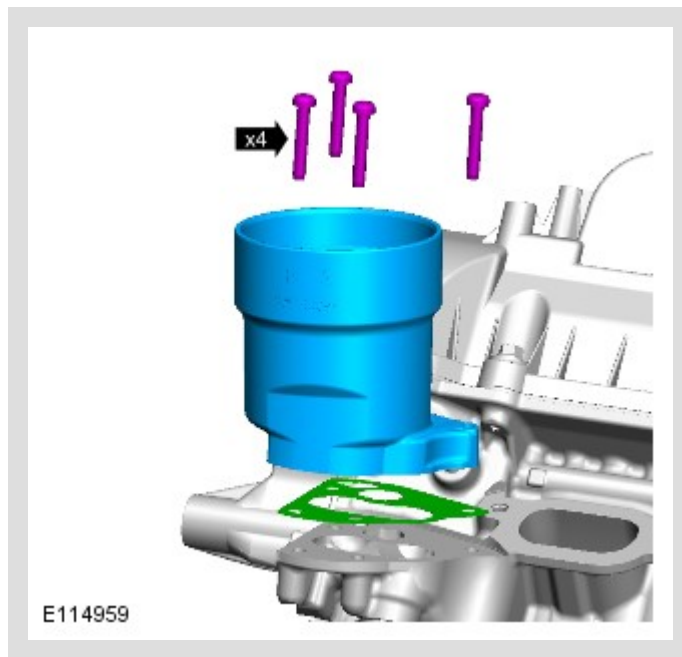
Remove and discard the O-ring seal.



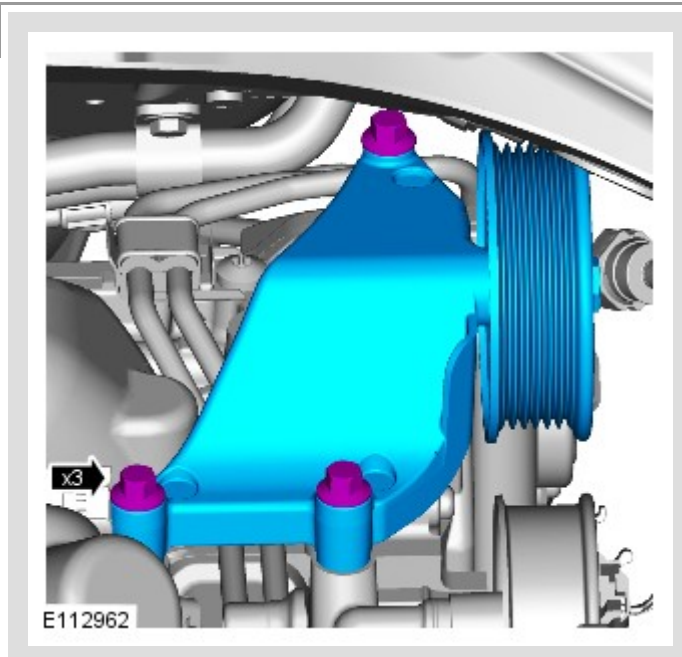
15.

 **NOTE:**

Discard the gasket.



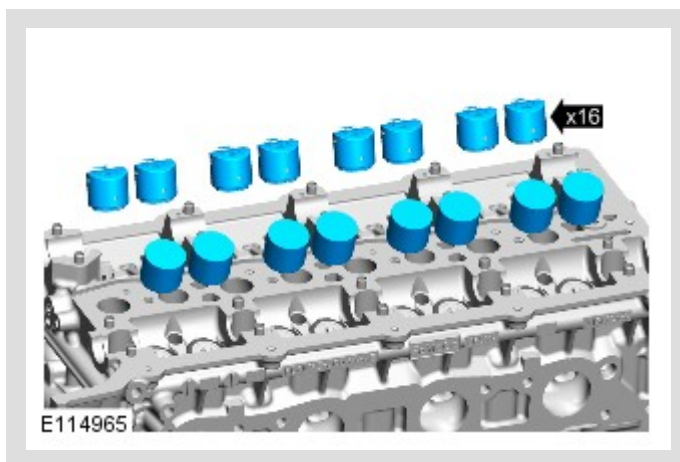
16.



17.

⚠ CAUTIONS:

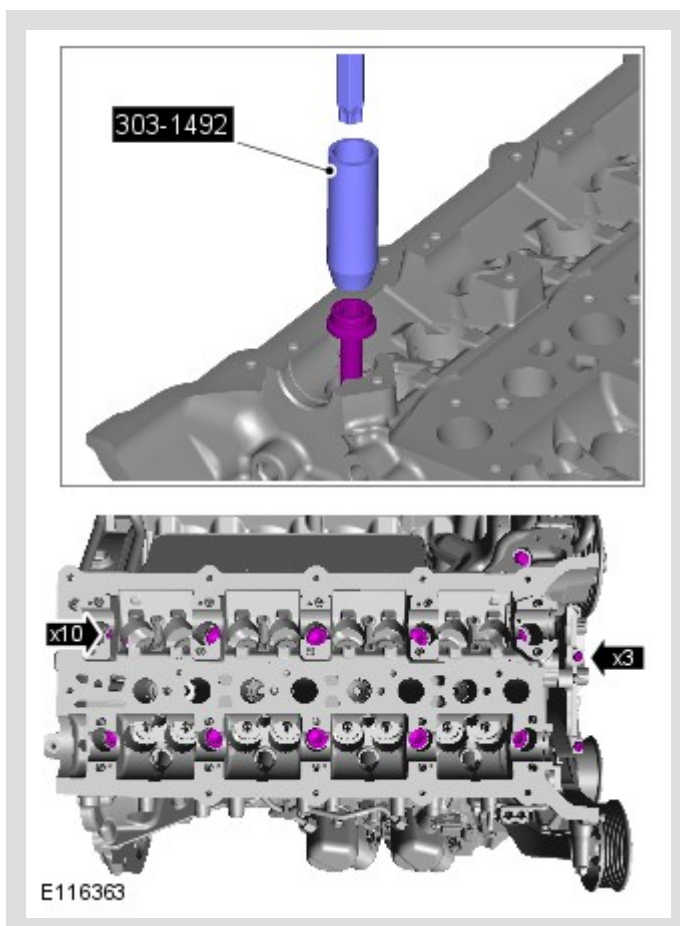
- If a new cylinder head has been installed then new tappets must be installed.
- If the cylinder head is being removed without a new component being installed, the tappets must be installed in their original positions.



18.

⚠ CAUTION:

Discard the bolts.

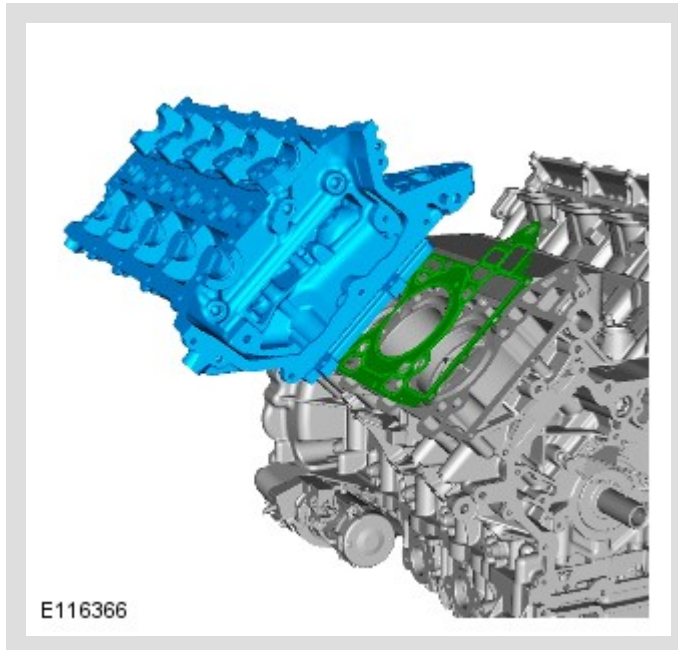


- *Special Tool(s):* 303-1492

19.

 NOTE:

Discard the gasket.

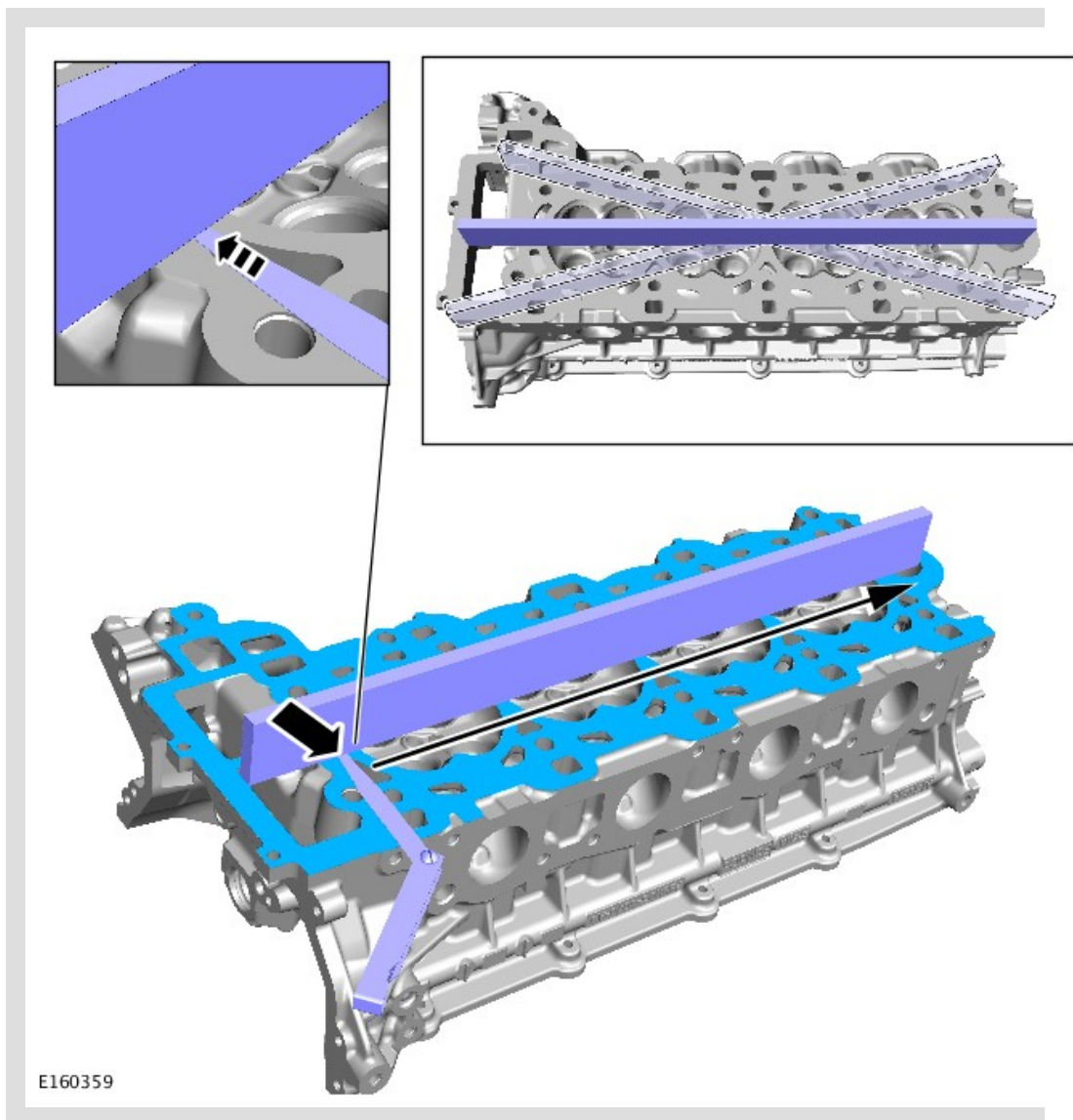


INSTALLATION

1.

 CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- An acceptable flatness of the cylinder head is 0.2mm.



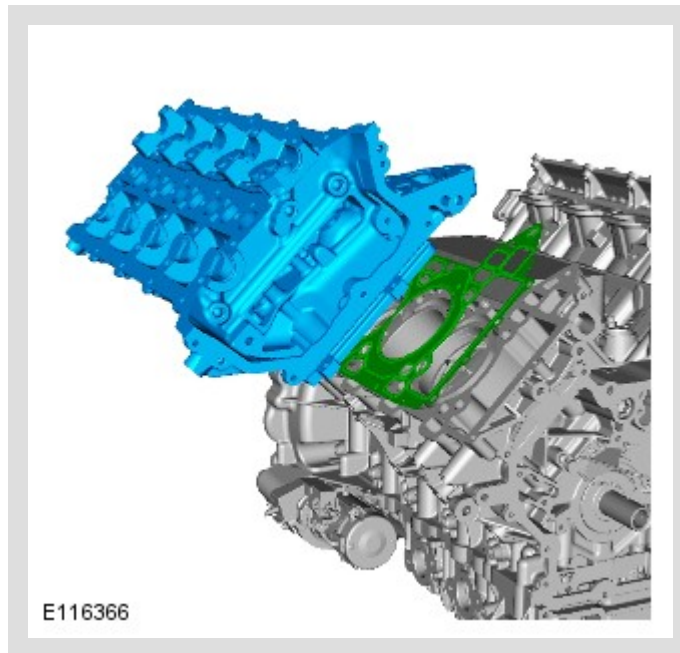
- Clean and inspect the cylinder head and cylinder block.
 - Check the cylinder head face for distortion, across the center and from corner to corner.
- Refer to: Cylinder Head Distortion (303-00 Engine System - General Information, General Procedures).



Make sure care is taken when handling the cylinder head gasket.

 NOTE:

Install a new gasket.

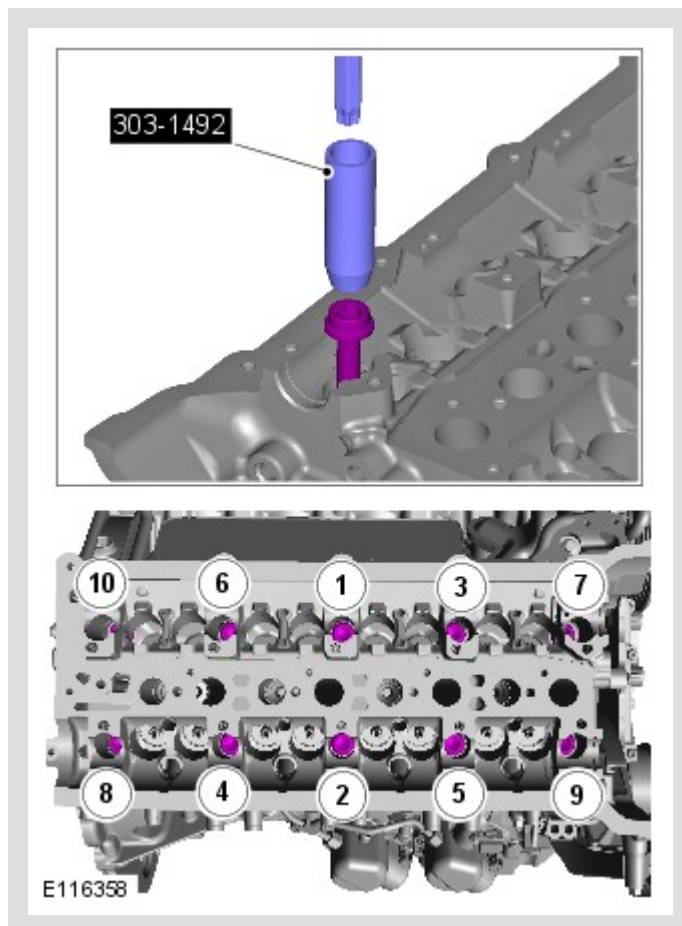


Renew Part: *Cylinder head gasket* Quantity: *1* .

3.

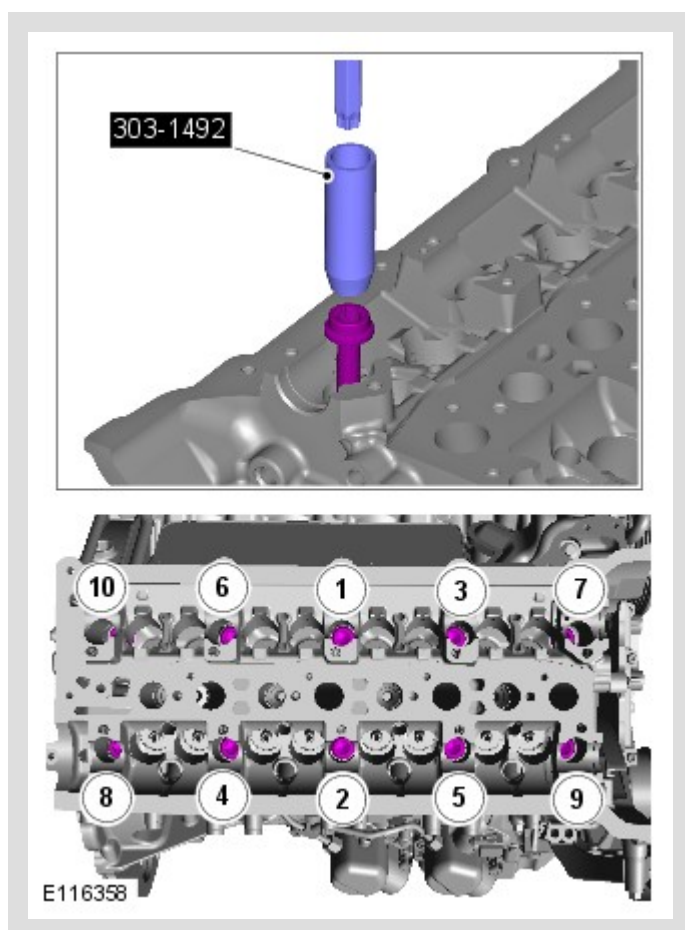
 CAUTION:

Make sure that new cylinder head bolts are installed.

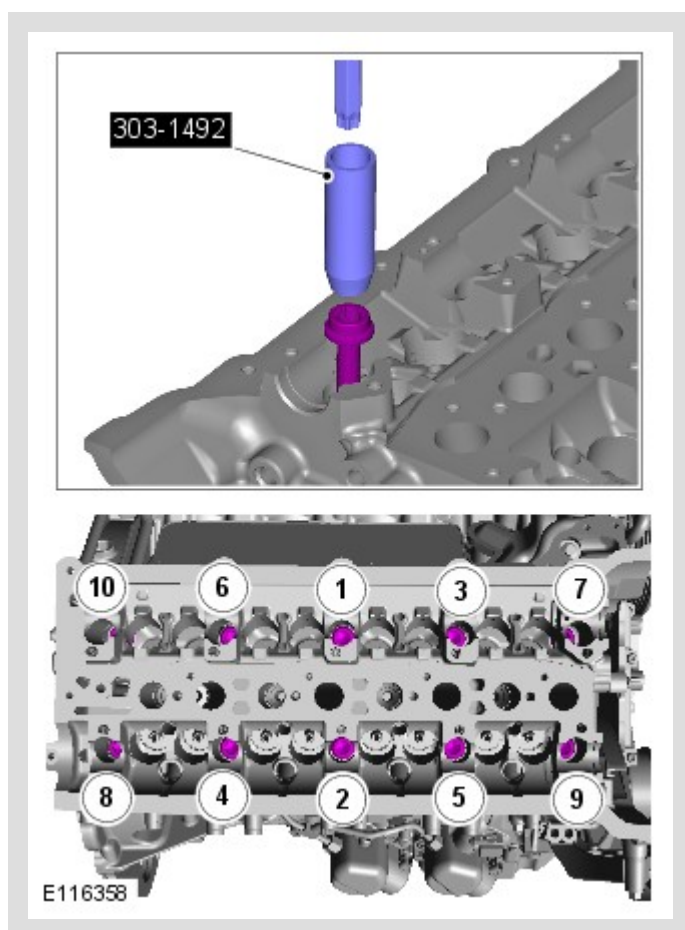


Renew Part: *Cylinder head bolts* Quantity: *1* .

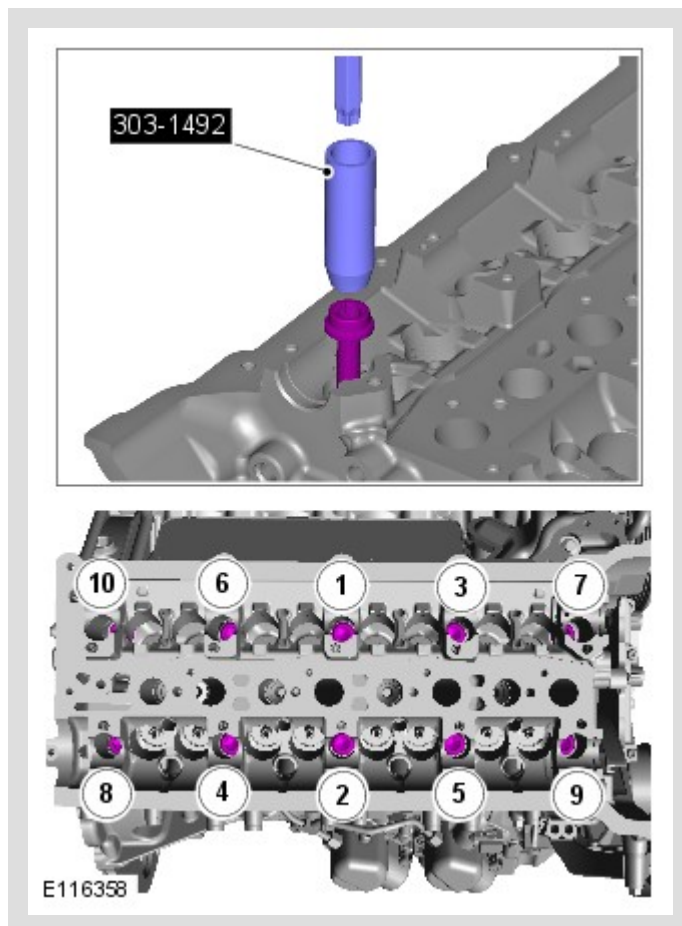
Torque: 20 Nm



Torque: 35 Nm

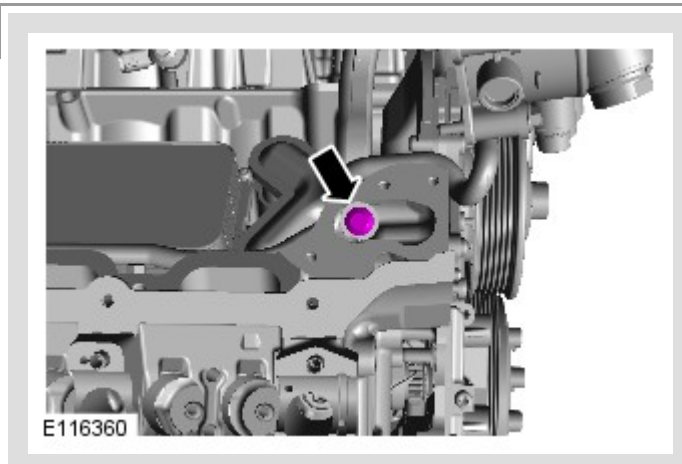


Tighten the bolts 1 to 10, a further 90 degrees.



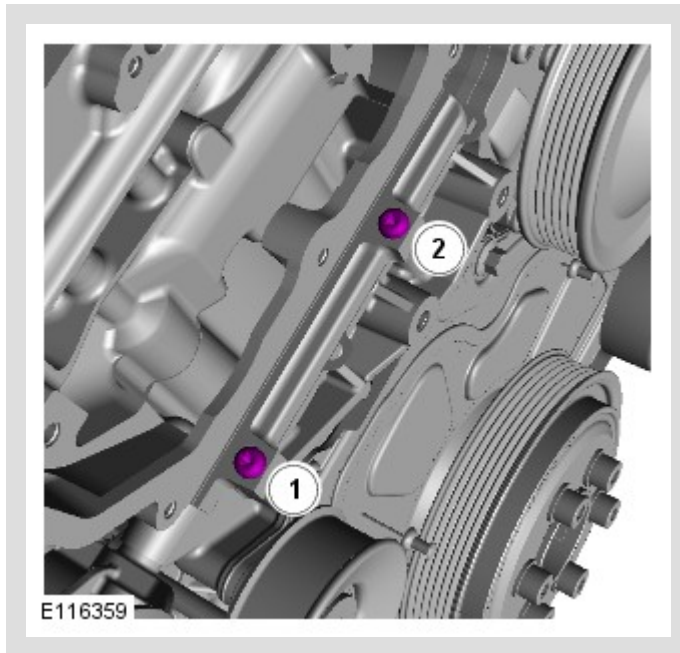
Tighten the bolts 1 to 10, a further 120 degrees.

7.



Torque: 25 Nm

8.



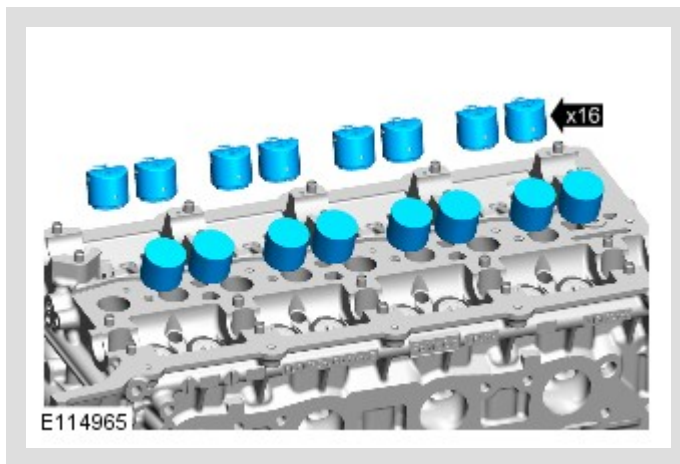
Torque: 12 Nm

9.



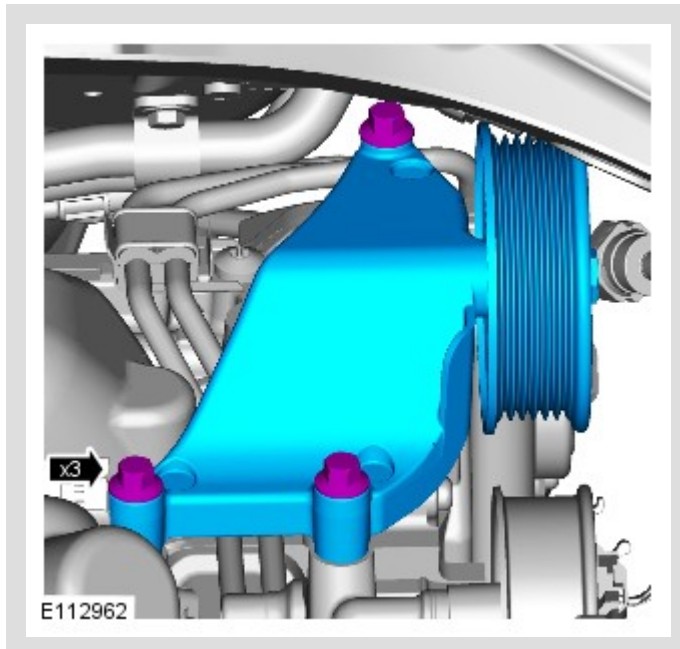
CAUTIONS:

- If a new cylinder head has been installed then new tappets must be installed.
- If the cylinder head is being removed without a new component being installed, the tappets must be installed in their original positions.



Lubricate the valve tappets with clean engine oil.

10.

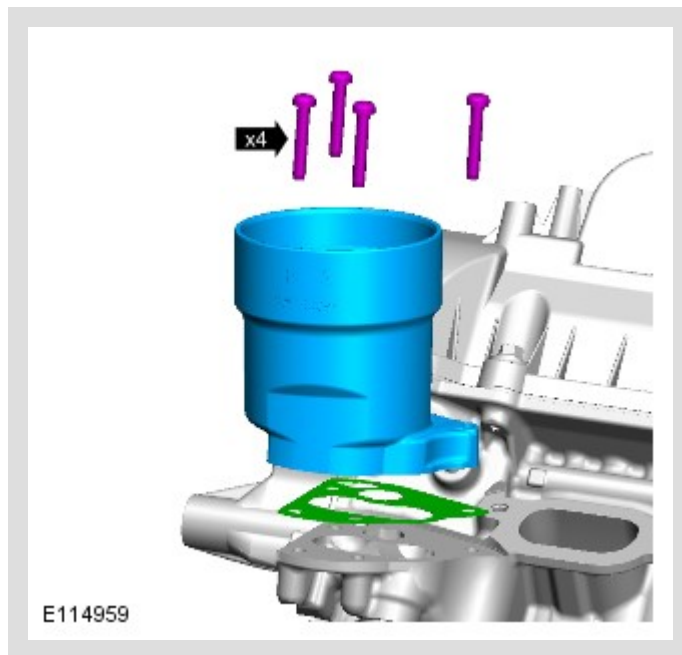


Torque: 25 Nm

11.

 **NOTE:**

Install a new gasket.



Renew Part: *Oil filter housing gasket* Quantity: *1* .

12.



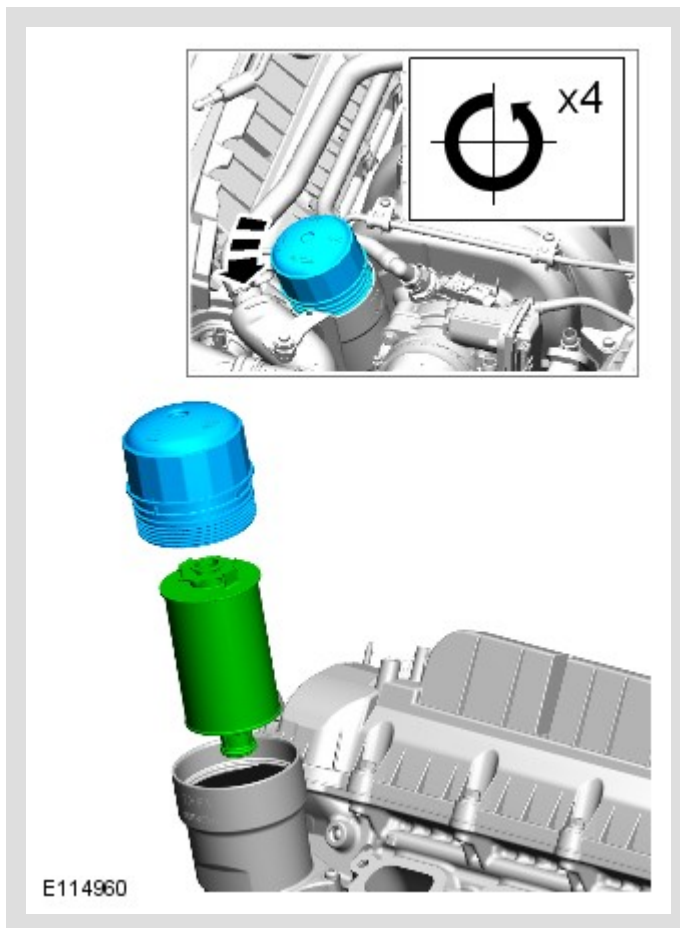
CAUTION:

A new O-ring seal is to be installed.



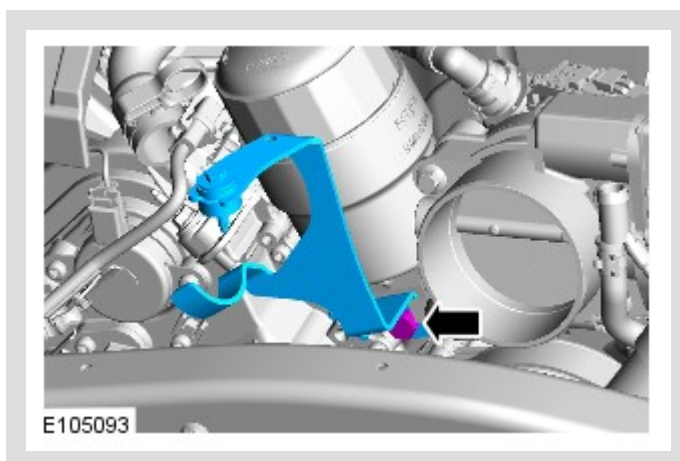
NOTE:

Install a new engine oil filter.



Renew Part: *Oil filter O-ring seal(s)* Quantity: 1 .

13.



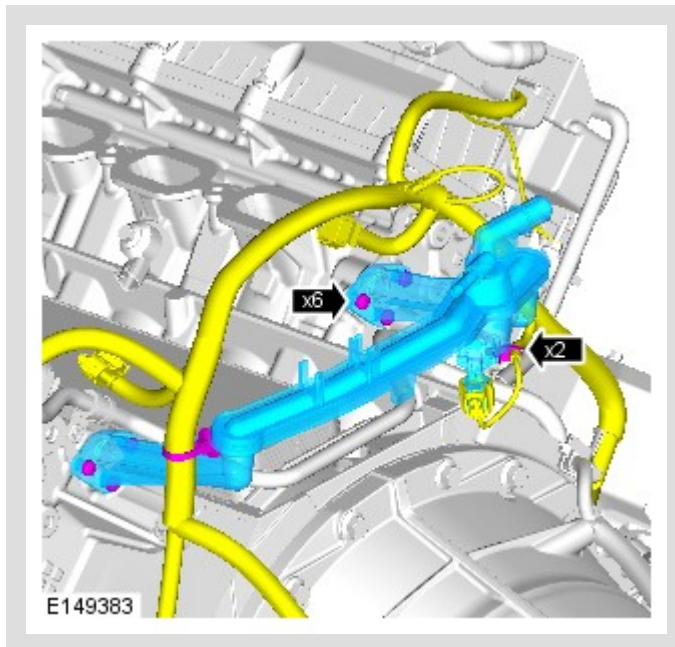
Torque: 25 Nm

14.



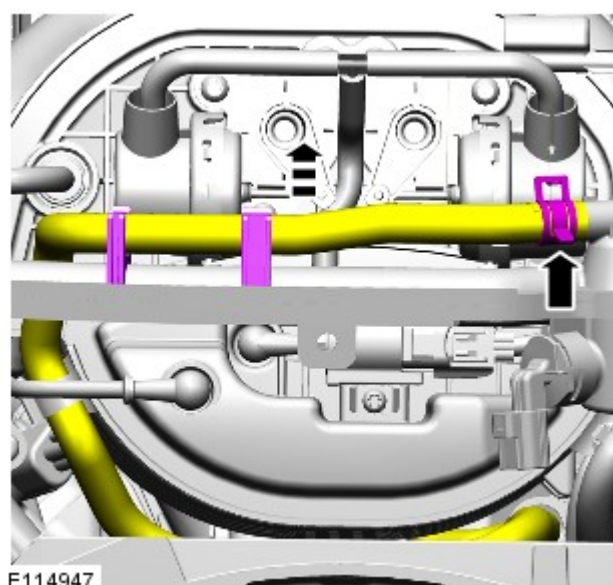
CAUTION:

Install the new seals.



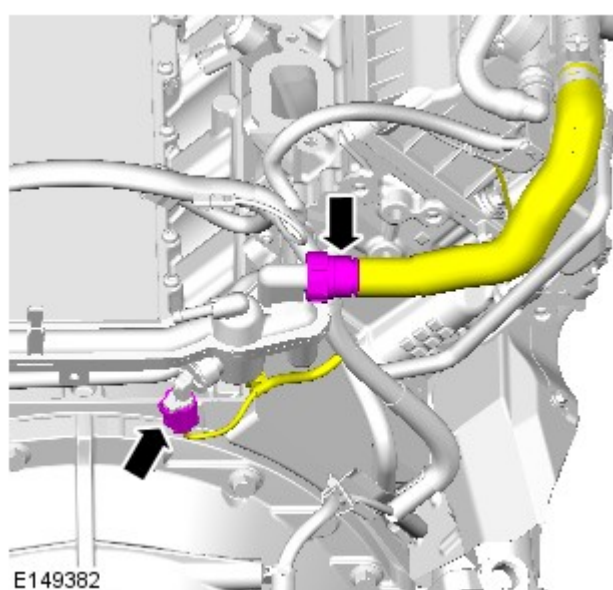
Renew Part: *Cylinder block coolant outlet O-ring seals* Quantity: 1 .
Torque: 10 Nm

15.



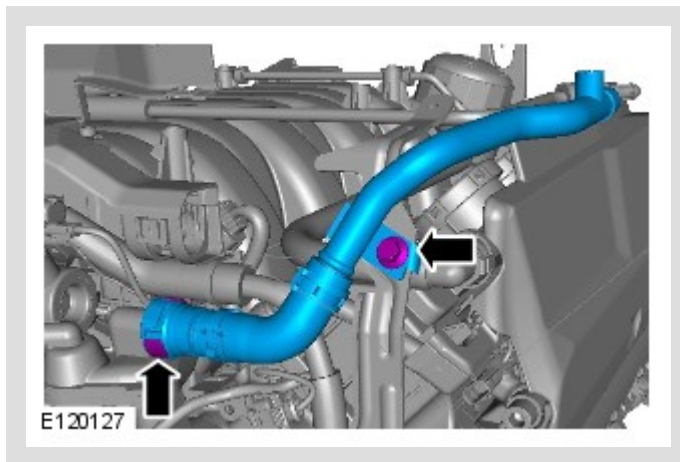
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16.



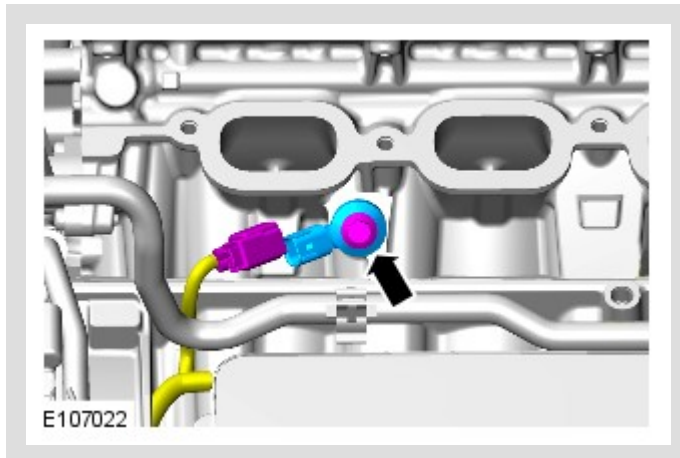
E149382

17.



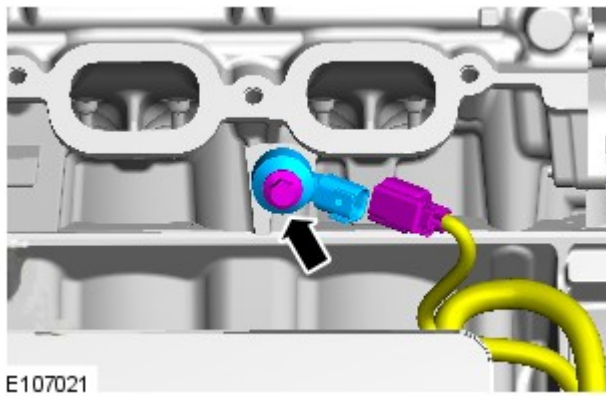
Torque: 10 Nm

18.



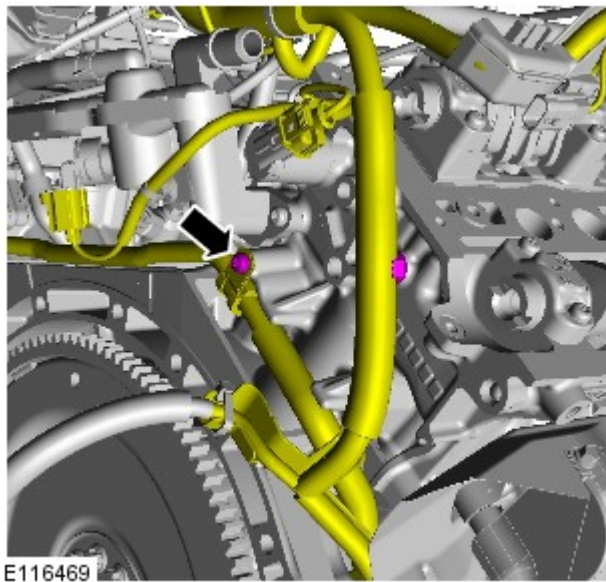
Torque: 20 Nm

19.



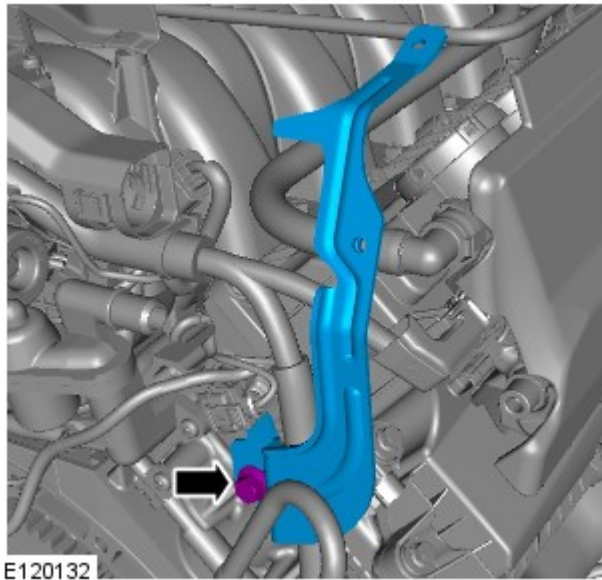
Torque: 20 Nm

20.



Torque: 12 Nm

21.



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22. Refer to: Right Exhaust Manifold (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).
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23. Refer to: Right Camshaft (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).
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24. Connect the battery ground cable.
Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).